Name ____ Date ___________ Period _____

Graphing f(x) = |x| and Key Attributes

TEACHER KEY | Algebra II Honors | Unit 2 | Day 1 | TEK A2.2.A

Objective: I can graph $f(x) = |x|$ from its piecewise definition, derive its key attributes algebraically, and compare its behavior to a linear function.

Vocabulary

Piecewise function — a function defined by **different rules** on different parts of the domain.

Absolute value, formally — $|x| = x$ if $x \geq 0$, and $|x| =$ **$-x$** if $x < 0$.

Even function — $f(-x) = f(x)$ for every x in the domain; the graph is symmetric about the **y-axis**.

Vertex — where the two pieces meet: **(0, 0)** for $f(x) = |x|$.

Part 1 — The Piecewise Definition

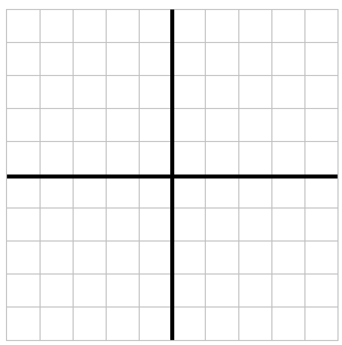
$f(x) = |x| = \quad x \;\text{ if } x \geq 0, \qquad -x \;\text{ if } x < 0$

Evaluate using the correct piece.

$f(-4) =$ **4** (uses -x) $f(6) =$ **6** (uses x) $f(0) =$ **0** $f(-0.5) =$ **0.5** (uses -x)

Part 2 — Graph Each Piece

Graph $y = x$ for $x \geq 0$ and $y = -x$ for $x < 0$ on the SAME grid - two rays meeting at the vertex. Each square is 1 unit, both axes run -5 to 5.



Part 3 — Key Attributes, Derived From the Definition

Domain: **$(-\infty, \infty)$** Vertex: **(0, 0)**

Derive the range from the two pieces, instead of just reading it off the graph:

For $x \geq 0$, the rule is $y = x$, so y takes every value **≥ 0**

For $x < 0$, the rule is $y = -x$, so y takes every value **> 0** (since x is negative, -x is positive)

Combining both pieces, the range is **$[0, \infty)$**

Now show algebraically that $f(x) = |x|$ is an even function:

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$f(-x) = |-x| =$ **$|x| = f(x)$** → confirmed even, so the graph is symmetric about the **y-axis**

Now derive the x- and y-intercepts from the definition instead of reading them off the graph:

x-piece: $x = 0$ → $x =$ **0** -x-piece: $-x = 0$ → $x =$ **0** (both agree)

So the only x-intercept is **(0, 0)**, which is also the y-intercept, since $f(0) =$ **0**

Part 4 — Compare and Explain

1. Explain why $f(x) = |x|$ has a minimum value but a line $y = mx + b$ $(m \neq 0)$ does not.

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   Sample: **the two pieces of $|x|$ both point upward from the vertex, so every output is ≥ 0 and 0 is the smallest one; a non-horizontal line keeps rising or falling forever in both directions**

2. For which values of x does $|x| = x$ (the two pieces agree)? **$x \geq 0$**

3. For which values of x does $|x| = -x$? **$x \leq 0$** (at $x = 0$ both pieces give the same value, 0)

4. Is $f(x) = |x|$ a function? Justify using the vertical line test.

   Yes **every vertical line crosses the graph exactly once**

Part 5 — Real-World: Distance, Justified

A hiker's position is x miles from a trailhead at 0 (negative = west, positive = east). Distance from the trailhead is $d(x) = |x|$.

Using the piecewise definition, explain why d(x) is never negative, no matter which direction the hiker walks.

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Sample: **if $x \geq 0$ (east), $d(x) = x \geq 0$; if $x < 0$ (west), $d(x) = -x$, and since x is negative, -x is positive - so both cases give $d(x) \geq 0$**

Part 6 — Going Further

1. For which value(s) of x does $|x| = 5$? **$x = 5$ or $x = -5$**

   Explain in one sentence why there are two answers.

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   Sample: **both 5 and -5 are a distance of 5 from zero**

2. For which value(s) of x does $|x| = -3$? **none - no solution**

   Explain why, using the range you derived in Part 3.

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   Sample: **the range of $|x|$ is $[0, \infty)$, so $|x|$ can never equal a negative number like -3**

3. Could $|x|$ ever equal a number strictly between -1 and 0, like -0.5? Explain.

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   No **same reasoning - the range is $[0, \infty)$, and -0.5 is negative**

Exit Ticket

Justify algebraically that $f(x) = |x|$ is symmetric about the y-axis, then give its domain, range, and vertex.

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Domain **$(-\infty, \infty)$** Range **$[0, \infty)$** Vertex **$(0, 0)$**